WEEK 3: SURFACE AREA OF SOLIDS (WITH ILLUSTRATIONS)

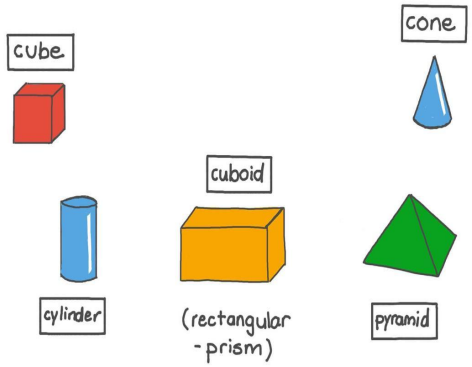
1. INTRODUCTION

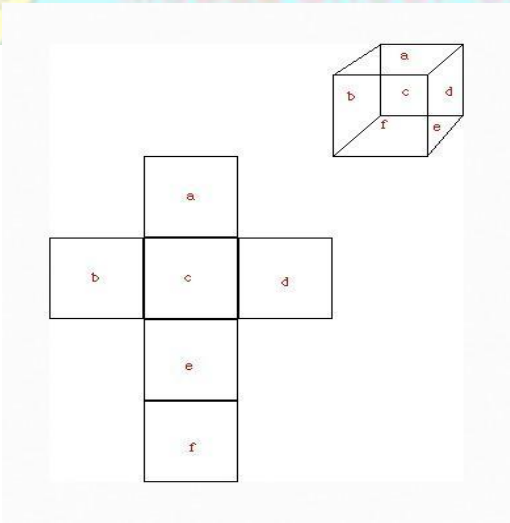
- Surface Area (SA) = total outer covering of a 3D shape
- It is the area of the **net (unfolded shape)**
- Measured in **cm², m²**

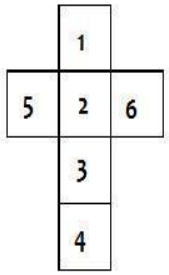
2. CONTENT

A. CUBE AND CUBOID

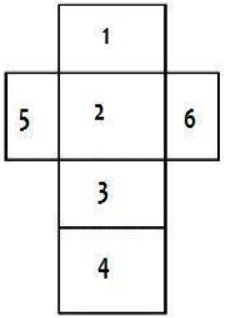
Cube







Net of cube



Net of cuboid

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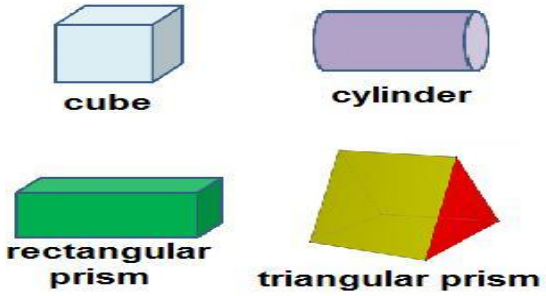
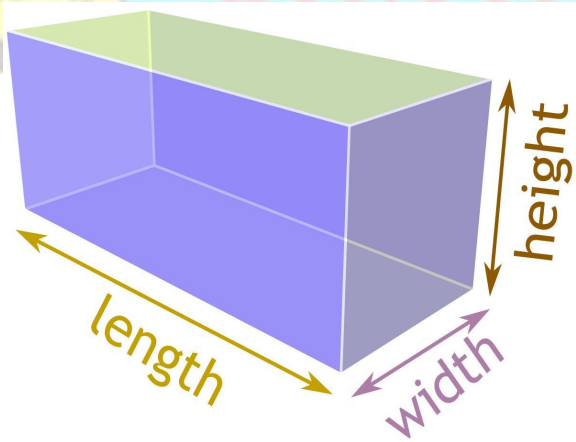
- All edges are equal

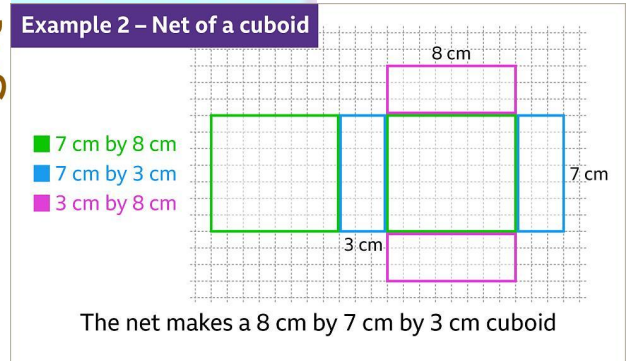
Formula:

- TSA = $6l^2$

Cuboid (Rectangular Prism)







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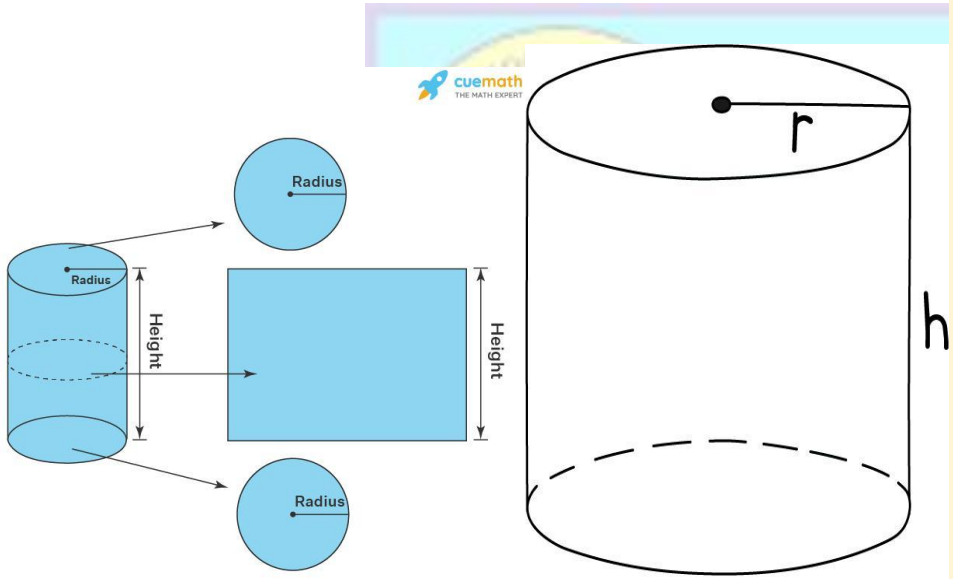
- Dimensions: length (l), width (w), height (h)

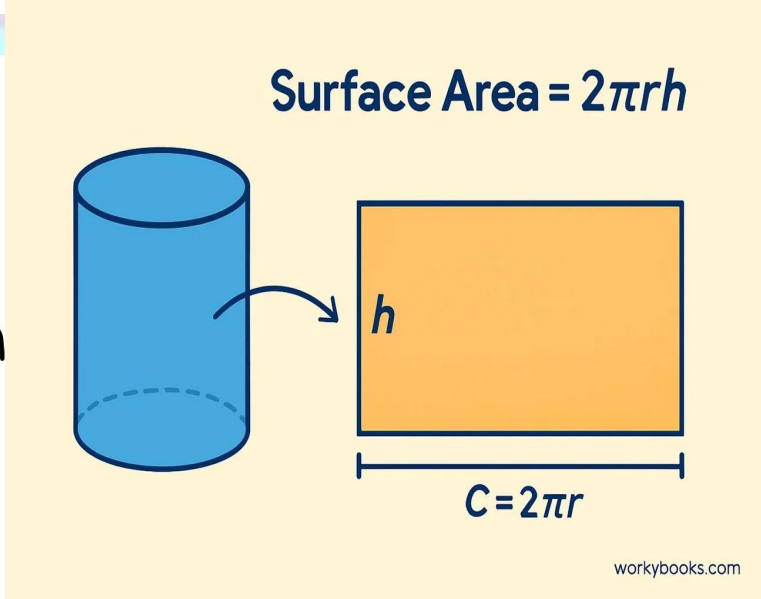
Formula:

- TSA = 2(lw + lh + wh)

B. CYLINDER AND CONE

Cylinder





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Components:

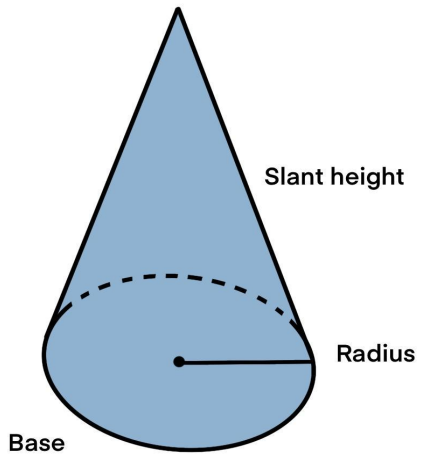
- 2 circular faces

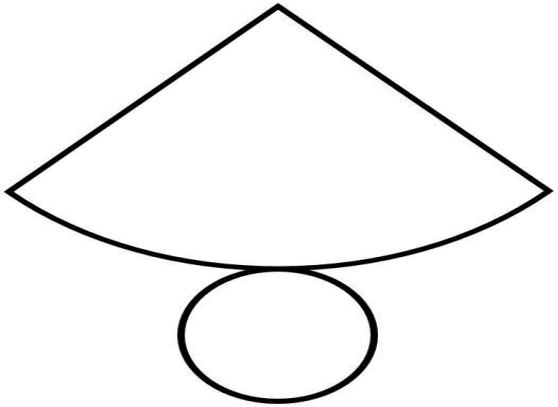
- 1 curved surface

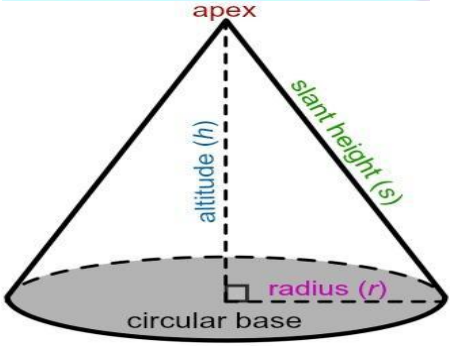
Formulas:

- CSA = $2\pi rh$
- TSA = $2\pi r^2 + 2\pi rh$
- TSA = $2\pi r(r + h)$

Cone







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Components:

- Base circle
- Curved surface

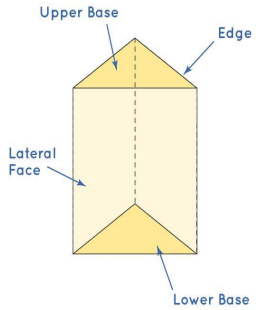
Formulas:

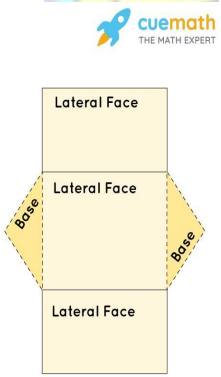
- CSA = πRl
- TSA = $\pi R^2 + \pi Rl$
- TSA = $\pi R(R + l)$

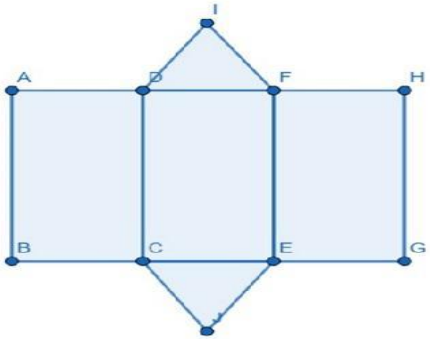
Key Relationship:

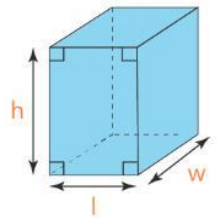
- $l^2 = h^2 + R^2$

C. PRISM

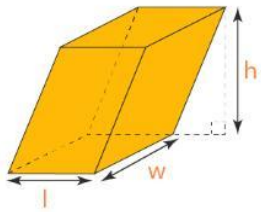








Right Rectangular Prism



Oblique Rectangular Prism

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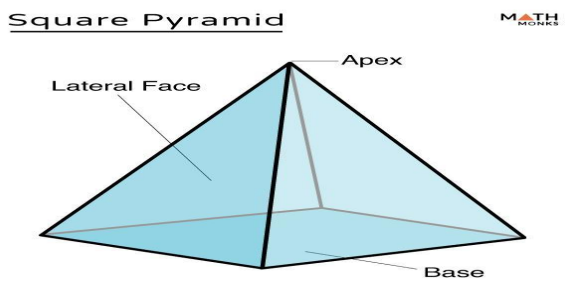
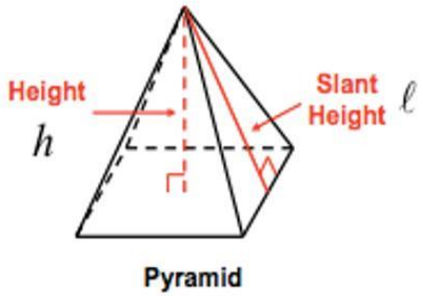
- Same cross-section throughout

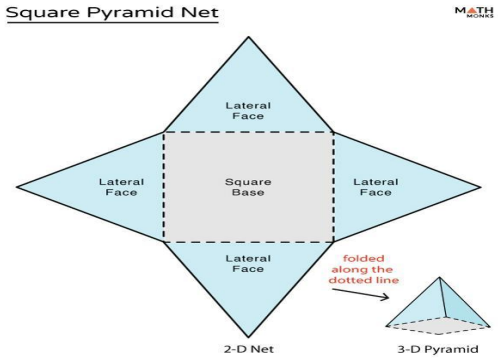
Formula:

- TSA = 2A + P × L

D. PYRAMID







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- Base + triangular faces meeting at apex

Formula:

- TSA = $l^2 + 2ls$

3. WORKED EXAMPLES (SHORT)

Example 1: Cuboid

l = 8, w = 5, h = 4

TSA = 2(lw + lh + wh)
= 2(40 + 32 + 20)
= **184 cm²**

Example 2: Cylinder

r = 7, h = 10

TSA = 2πr(r + h)
= 44 × 17
= **748 cm²**

Example 3: Cone

R = 5, h = 12

$l^2 = 12^2 + 5^2 = 169$
l = 13

TSA = πR(R + l)
= π × 5 × 18
= **90π cm²**

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4. REVISION QUESTIONS

1. Define surface area
2. Formula for cube SA
3. Parts of a cylinder
4. Difference between h and l
5. State $l^2 = h^2 + R^2$

QUESTION (a): PYRAMID ON CUBOID – VOLUME

Question

The diagram shows a pyramid standing on a cuboid. The dimensions of the cuboid are:

- Length = 4 m
- Width = 3 m
- Height = 2 m

The slant edge of the pyramid is 5 m.
Calculate the **volume of the whole shape**.

STEP 1: UNDERSTANDING THE SHAPE

Total Volume =
Volume of cuboid + Volume of pyramid

STEP 2: VOLUME OF CUBOID

Formula:

- Volume = $l \times w \times h$

Substitute:

- Volume = $4 \times 3 \times 2$
- Volume = $24\ m^3$

STEP 3: FIND HEIGHT OF PYRAMID

We need vertical height of pyramid (h).

The pyramid sits on a rectangular base 4 m × 3 m.

Half dimensions:

- Half length = 4 ÷ 2 = 2 m
- Half width = 3 ÷ 2 = 1.5 m

Using Pythagoras:

slant edge² = height² + (distance from center to corner)²

Distance from center to corner:

$= \sqrt{(2^2 + 1.5^2)}$
$= \sqrt{(4 + 2.25)}$
$= \sqrt{6.25}$
= 2.5 m

Now:

$5^2 = h^2 + 2.5^2$
$25 = h^2 + 6.25$
$h^2 = 18.75$
h = 4.33 m (approx)

STEP 4: VOLUME OF PYRAMID

Formula:

- Volume = 1/3 × base area × height

Base area:

- $4 \times 3 = 12\ m^2$

So:

- Volume = $1/3 \times 12 \times 4.33$
- Volume = 4×4.33
- Volume = $17.32\ m^3$

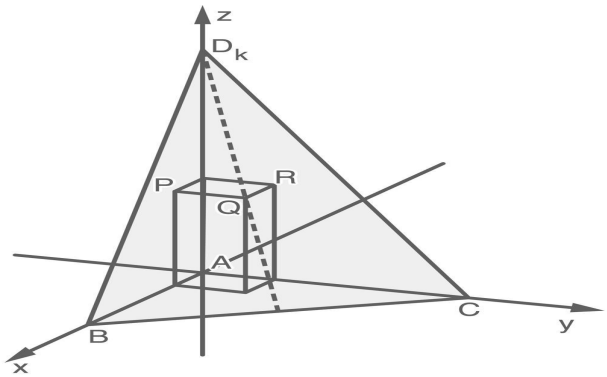
STEP 5: TOTAL VOLUME

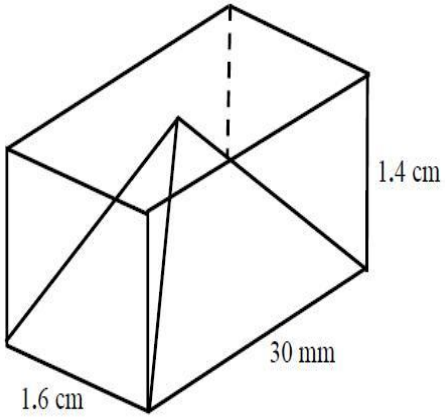
Total Volume = cuboid + pyramid
= 24 + 17.32
= **41.32 m³**

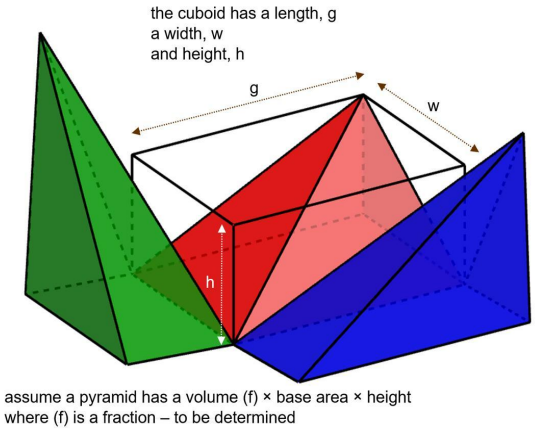
FINAL ANSWER (a):

✔ **Volume = 41.32 m³**

DIAGRAM (ILLUSTRATION)







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5. ASSESSMENT

IN-CLASS

- Write formulas for:
  - Cuboid TSA
  - Cylinder CSA
  - Cone TSA

ASSIGNMENT

1. Room dimensions → find wall + ceiling area
2. Cylinder → find height and TSA

3. Pyramid → find TSA
4. Cone → find height and TSA

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